



ETH*zürich*

Product Development of a Substrate Mixer and Portioning Device

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CONTENTS

List of Tables	vi
List of Figures	vii
Abstract	viii
1 Introduction	1
1.1 Economic and Ecological Significance of Mushroom Cultivation	1
1.2 The Technical Process of Mushroom Cultivation	1
1.2.1 Industrial Implementation of Substrate Preparation	2
1.3 Mushroom Cultivation in Developing Regions	3
1.3.1 Smallholder Constraints in Kampala, Uganda	3
1.4 Justification and Research Questions	3
2 Methods	5
2.1 Methodology and Technical Design Approach	5
2.1.1 Experimental Strategy and Data Management	5
2.1.2 Foundational Design Principles	6
2.2 Product Development	6
2.2.1 Material Selection	6
2.2.2 Mixing Vessel and Supporting Structure	7
2.2.3 Mixing: Agitator Assembly	8
2.2.4 Watering: Manifold System	10
2.2.5 Portioning: Filling Line	11
3 Results and discussion	14
3.1 Deliverables and Operational Throughput	14
3.2 Evaluation of Substrate Homogeneity (RQ1)	14
3.2.1 Water Content in Bags	14
3.2.2 Longitudinal Transport Efficiency of Agitator	15

3.3	Evaluation of Uniform Hydration (RQ2)	16
3.3.1	Manifold Flow Distribution	16
3.3.2	Water content while Mixing	17
3.4	Evaluation of Portioning Accuracy (RQ3)	18
4	Conclusions	19
4.1	Prototype Performance and Research Questions	19
4.2	Limitations and System Boundaries	19
4.3	Recommendations for Future Engineering Work	20
	Bibliography	21
	Appendix A - Test Reports from Zero-Series Experiments	24
	Appendix B - Data Management for First-Series Experiments	25
	Appendix C - Technical Drawings, User Manual and Building Instructions	27
	Appendix D - Correspondences, Industrial Field Visits and Expert Interviews	28

LIST OF TABLES

3.1 Portioning performance summary	18
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LIST OF FIGURES

2.1	Fully assembled final prototype.	7
2.2	Model and final agitator configuration.	9
2.3	Final version of filling line and visualized substrate blockages.	12
3.1	Longitudinal transport efficiency of the agitator under non-uniform initial loading profiles.	15
3.2	Manifold flow distribution.	16
3.3	Substrate moisture content profiles recorded across successive mixing intervals.	17

ABSTRACT

Mushroom cultivation supports the circular economy in developing regions. In Uganda, mushroom substrate preparation is mostly manual and causes severe labor constraints and inconsistent process quality. This thesis details the engineering design and quantitative evaluation of a manually operated substrate mixing and portioning device for smallholders in Kampala, Uganda. The objective was to build a low-cost, low-tech, and locally reproducible apparatus. The system integrates substrate homogenization, hydration, and volumetric dosing. Prototype testing demonstrated high blending homogeneity and rapid material transport. Post-processing analysis showed an average absolute deviation in water content of $(0.45 \pm 0.06)\%$, confirming precise moisture regulation. Furthermore, the portioning mechanism kept mass deviations below the $\pm 10\text{ wt}\%$ tolerance threshold. These findings demonstrate that hand-driven mechanical solutions can successfully standardize substrate processing. The system reduces labor intensity and increases consistency in decentralized mushroom cultivation.

1 INTRODUCTION

1.1 Economic and Ecological Significance of Mushroom Cultivation

Within the circular economy, mushroom cultivation serves as an effective mechanism for waste valorization by using agricultural byproducts, such as straw, sawdust, cotton lint, or coffee husks, that are otherwise discarded as organic waste ([Grimm & Wösten, 2018](#); [Oei & van Nieuwenhuijzen, 2005](#)). This bioconversion process mitigates environmental pollution by transforming low-value residues into high-quality food and potent organic fertilizers ([Grimm & Wösten, 2018](#)). Furthermore, this model facilitates waste-stream monetization, effectively transforming discarded residues into marketable assets ([Ramirez et al., 2021](#)).

Due to their high yield potential and ease of cultivation, *Pleurotus* mushroom species valorize various residues, mitigating the environmental and health hazards of indiscriminate dumping or open burning ([Dissasa, 2022](#)). For example, raw coffee husks are ecologically liable due to ingredients toxic to microorganisms and unsuitable for livestock feed ([Dissasa, 2022](#)). Cultivating oyster mushrooms transforms a hazardous waste stream into a high-yield commodity ([Dissasa, 2022](#)). Similarly, cellulosic industrial byproducts, such as cotton gin trash, are diverted from environmentally harmful landfill streams to serve as structural substrate foundations ([Grimm & Wösten, 2018](#)).

Additionally, the high nutritional density of mushrooms ([Oei, 2003](#), pp. 4-5) provides a critical supplement to staple-based diets in resource-constrained environments, supporting localized interventions against malnutrition ([Akter et al., 2022](#)).

1.2 The Technical Process of Mushroom Cultivation

The technical process of cultivation begins with substrate preparation, where agricultural residues must be comminuted, hydrated to an optimal target content, and thoroughly mixed. The primary objective of mixing is to achieve the most uniform distribution of substrate compo-

nents and moisture possible. This is particularly critical when incorporating micro-ingredients, such as chalk; localized nutrient concentrations or "lumps" can foster competitive microbial growth and lead to contamination. Furthermore, precise hydration is required to ensure a uniform moisture content of 60–70% throughout the substrate (Oei & van Nieuwenhuijzen, 2005, p. 38).

Following preparation, the substrate is filled into bags and pasteurized or sterilized to suppress competitive microorganisms (Oei, 2003, pp. 19-26). Only once the substrate has cooled to ambient temperature is it inoculated with mycelium pre-grown on grains, referred to as "spawn." After an incubation period of approximately two to six weeks, the mycelium fully colonizes the substrate. The transition to the generative phase and the emergence of fruiting bodies are subsequently induced by specific environmental stimuli, such as thermal fluctuations or physical agitation (Oei & van Nieuwenhuijzen, 2005, p. 61). Upon reaching physiological maturity, the harvested fruiting bodies enter the commercial value chain as high-value edible commodities, fulfilling both the nutritional and economic objectives of the production cycle.

1.2.1 Industrial Implementation of Substrate Preparation

In industrialized contexts, substrate preparation has evolved into a highly mechanized, often empirically-driven engineering discipline. Because optimal parameters for various mushroom species are frequently determined through iterative local experimentation (Wigger & Fanger, 2026), industrial facilities exhibit a wide diversity in mixing technologies. Standard configurations often include ribbon mixers, rotary drum mixers (Oei, 2003, pp. 22-23), or paddle mixers (Joconn Machinery Co., n.d.). These systems are utilized for their proven ability to effectively homogenize components of varying particle sizes while ensuring that water is integrated uniformly across the material matrix (Oei, 2003, pp. 22-23). As the substrate forms the fundamental basis of cultivation, inconsistencies in the initial mixing or hydration stages directly correlate with diminished or variable yields, rendering the substrate preparation phase a critical step for production success (Suwannarach *et al.*, 2022).

Evidence from large-scale commercial operations in Switzerland indicates a prevalent strategy of repurposing heavy-duty machinery from the livestock and food sectors. For instance, vertical cone mixers achieve high-volume homogenization through internal screw systems (Schneider, 2026). Furthermore, state-of-the-art production lines prioritize process intensification by integrating multiple unit operations like mixing, hydration, and pasteurization into one single vessel in an automated sequence, followed by mechanized bagging (Schneider, 2026; Wigger & Fanger, 2026). This integration reduces manual labor and significantly mitigates the risk of substrate contamination during material transfer (Wigger & Fanger, 2026).

1.3 Mushroom Cultivation in Developing Regions

Mushroom farming is a highly viable agricultural strategy for low-income countries, offering high financial yield per unit area with minimal land requirements (Khan *et al.*, 2024; Ramirez *et al.*, 2021). The utilization of locally sourced waste reduces reliance on external inputs, aligning with the constraints of resource-limited settings (Higgins *et al.*, 2017).

However, existing industrial mechanization is often precluded by high capital requirements and incompatibility with small-scale infrastructure (Higgins *et al.*, 2017). Additionally, the unreliable and expensive energy supply in countries such as Uganda (Mbabazi *et al.*, 2013) requires a manual approach to production. This dependence on manual labor introduces significant technical challenges: heterogeneous particle sizes make uniform mixing difficult, and slow water uptake by dry substrates leads to inconsistent moisture levels that jeopardize sterilization (Oei, 2003, pp. 89-90). Similarly, achieving uniform portioning and bag density, critical for consistent yields, is also difficult to maintain manually (Suwannarach *et al.*, 2022).

1.3.1 Smallholder Constraints in Kampala, Uganda

In Uganda, the market demand for mushrooms significantly exceeds the current supply (Kisseka, 2026). Nevertheless, production is constrained by intensive manual labor requirements. As indicated by Kisseka (2026) from *Hello Mushrooms U Ltd* in Kampala, these labor inputs represent the main barrier to scaling smallholder operations. Currently, the mixing, moistening, and bagging processes are performed manually, which is physically demanding and results in prohibitive labor costs (Kisseka, 2026).

A primary objective of this project is to significantly reduce operational man-hours and personnel requirements for substrate processing. Furthermore, the transition to a mechanized workflow aims to standardize production through system-inherent parameters. Using a device with defined operational procedures, the resulting substrate bags achieve more consistent weights, uniform moisture content, and homogeneous compositions. This approach aims at high-quality output without relying on the constant manual diligence of workers, providing a solid foundation for increased production capacity.

1.4 Justification and Research Questions

Small-scale producers in Uganda remain limited by the technical inconsistencies of manual substrate preparation. Current practices are labor-intensive and prone to variations in composition and moisture content, which increase contamination risks and reduce process reliability.

Industrialized solutions remain economically and technically inaccessible. This necessitates the development of a low-cost, robust, and locally reproducible device that integrates mixing, hydration, and portioning. By providing open-source construction documentation, this project aims to enable smallholders in Kampala to improve production efficiency and secure sustainable income through standardized processes.

This work addresses the design and evaluation of a hand-driven substrate mixer and portioning device. The engineering challenges are divided into three functional domains, *mixing*, *watering*, and *portioning*, which form the basis for the following research questions:

1. **Mixing (RQ1):** How can heterogeneous agricultural residues be processed to achieve a homogeneous substrate composition throughout the mixing vessel?
2. **Watering (RQ2):** How can the substrate be uniformly hydrated to achieve a target moisture content of 60–70 wt%?
3. **Portioning (RQ3):** How can the processed substrate be portioned into bags with deviations within a tolerance of ± 10 wt%?

These questions establish the functional requirements and provide measurable criteria for the evaluation of the developed prototype, bridging the practical constraints with the engineering design approach detailed in the subsequent chapter.

2 METHODS

2.1 Methodology and Technical Design Approach

This project follows a product development framework to address the technical challenges of substrate preparation in resource-constrained environments. The overarching research design is a descriptive and iterative engineering study focused on the realization and validation of a functional prototype. This approach is chosen to systematically address the three main research questions regarding uniform mixing, controlled hydration, and precise portioning as discussed in section 1.4.

To manage technical risks, a hybrid development model is employed. Subsystems with well-defined requirements are developed through a linear process, while high-uncertainty components are refined through iterative prototyping according to [von Petersdorff-Campen \(2024a\)](#). This process systematically isolates technical bottlenecks and validates different solutions through empirical metrics ([von Petersdorff-Campen, 2024b](#)).

The following subsections detail the execution of this methodology, beginning with the foundational design principles (2.1.2) and material selection criteria (2.2.1). Subsequent sections describe the specific development and testing of the supporting structure and mixing vessel (2.2.2), the agitator assembly (2.2.3), the water distribution system (2.2.4), and the portioning unit (2.2.5).

2.1.1 Experimental Strategy and Data Management

The experimental evaluation follows an agile, evolutionary prototyping framework divided into two distinct phases: formative Zero-Series experiments and summative First-Series evaluations. The Zero-Series comprised early-stage component testing and served as immediate formative engineering input that directly dictated subsequent design modifications. Their qualitative and structural outcomes are integrated directly into the technical descriptions in Sections 2.2.2 to 2.2.5 to justify the final prototype configuration.

In contrast, Sections 3.2 to 3.4 focus exclusively on the First-Series, which provides a system-

atic, quantitative evaluation of the final full-scale prototype. This phase assesses performance metrics that include substrate moisture distribution, portioning homogeneity, water flow consistency, and longitudinal transport efficiency. To ensure transparency and adherence to international research data management standards, these quantitative datasets are managed via a reproducible script-based computational processing pipeline. Detailed documentation regarding the execution of these experiments and the corresponding dataset structure is archived in `README.md` on [Google Drive](#) or [GitHub](#).

2.1.2 Foundational Design Principles

The development of the substrate mixer and portioning device is governed by a set of foundational principles tailored to the socioeconomic and technological context of smallholder farmers in Uganda. To ensure that the equipment remains economically accessible and locally maintainable, a strict low-cost and low-tech paradigm is adopted. As discussed in section 1.3, the device is designed for strictly manual, hand-driven operation.

The design utilizes universally available and standardized materials, simple geometries, and manufacturing techniques commonly found in conventional construction or foundational crafting. This approach guarantees that the open-source device can be easily reproduced and repaired using local infrastructure.

2.2 Product Development

2.2.1 Material Selection

Timber and steel were selected as the main construction materials due to their global availability, low cost, and ease of processing with basic tools. From an engineering perspective, wood and steel offer a good tradeoff between specific modulus E/ρ and specific strength σ_f/ρ (Ashby, 2011, p. 71). Therefore, wood and steel exhibit excellent performance in terms of stiffness and strength.

Especially wood makes an ideal, sustainable and affordable alternative to metals for static load-bearing frames (Ashby, 2011, p. 94). Therefore, Three-ply wood panels were used for the supporting structure and several parts of the filling line and water distribution system.

The agitator's assembly comprises standardized steel pipes and reinforcing bars, both of which are accessible on the Ugandan market (Kisseka, 2026). Steel is also selected for its high strength-to-weight ratio (Ashby, 2011, p. 71) and is therefore able to withstand the torque and mechanical stress induced during the manual agitation of high-density, moistened substrates without sustaining permanent deformation.

All components in direct contact with the moistened substrate require surface treatments to prevent material degradation (Bender & Göhlich, 2020a, pp. 691-724). The hygroscopic wooden elements are lacquered to resist moisture absorption and subsequent swelling (Burmester, 1970, p. 36). Similarly, the steel pipes and reinforcing bars of the agitator are coated with rust protection.

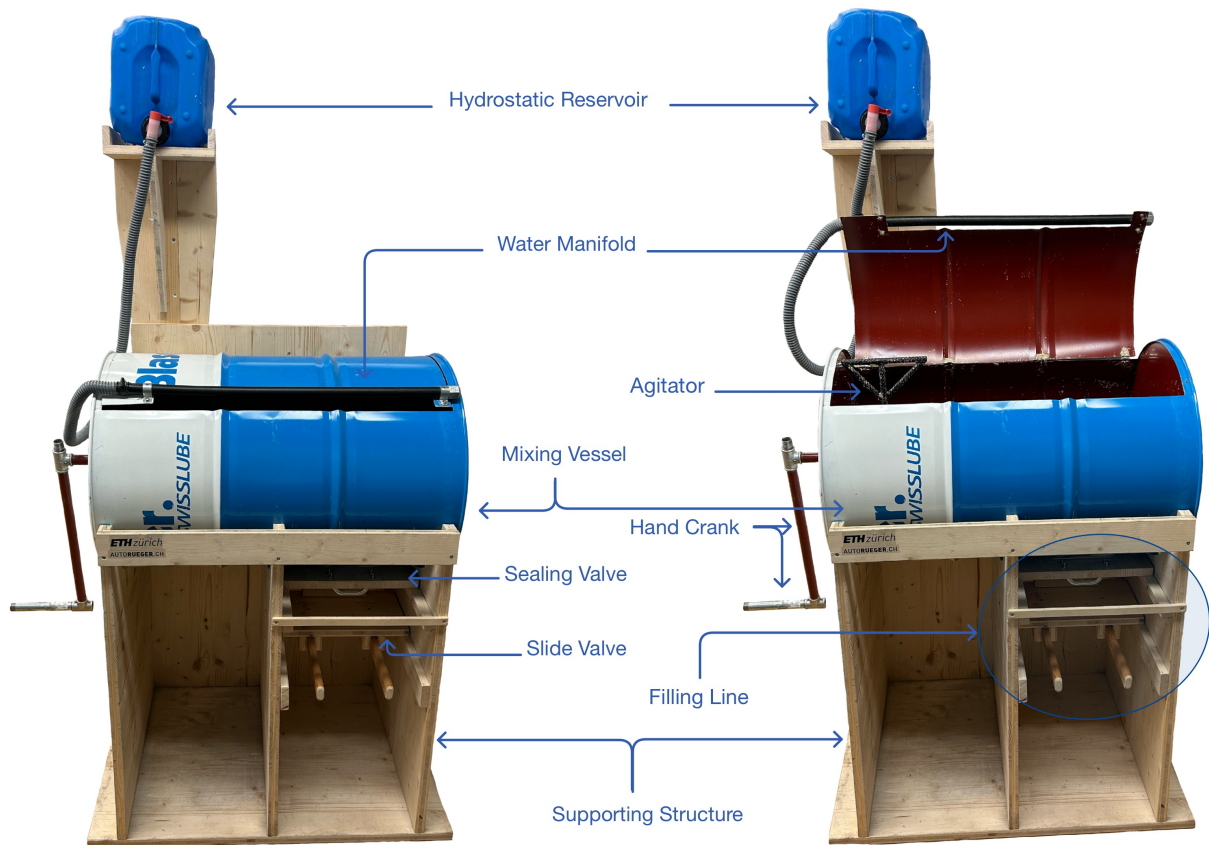


Figure 2.1. Fully assembled final prototype.
Closed lid (left) and open lid (right).

2.2.2 Mixing Vessel and Supporting Structure

The development of the foundational hardware which comprises the mixing vessel and its supporting frame is governed by the requirement to provide a stable, ergonomic and durable interface for the substrate preparation process. To ensure operational safety and process efficiency, the assembly must satisfy several technical conditions: it must support high static loads from the substrate and agitator assembly, neutralize dynamic forces such as torque and mass imbalances during mixing, and maintain an ergonomically optimal working height at approximately the operator's elbow level (Castellucci *et al.*, 2020).

Mixing Vessel

The mixing vessel was selected from standardized industrial containers. A 210-liter steel drum according to [ISO \(2002\) 15750](#) was chosen over alternative geometries such as the cubic Intermediate Bulk Container (IBC, [ISO \(2003\) 16487](#)). This choice is based on the cylindrical geometry of the drum, which facilitates a uniform radial flow and prevents the formation of so-called "dead zones", stagnant areas of unmixed substrate that would otherwise compromise homogeneity. The drum is oriented horizontally with its longitudinal axis parallel to the ground to prevent gravitational moisture gradients, which occur in vertical configurations. To allow for the installation of an internal agitator that spans the full diameter of the vessel, the drum is sectioned longitudinally across its entire width, as seen in figure [2.1](#).

Supporting Structure

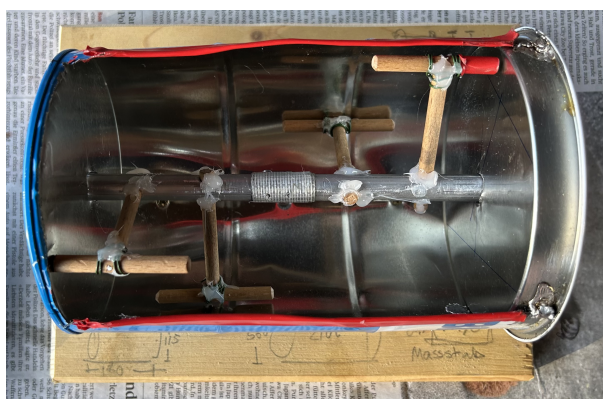
The supporting structure transitions from an initial beam-based design to a high-stiffness panel construction. Wooden beams were discarded due to insufficient lateral stability and interface complexity with the cylindrical vessel. The final architecture utilizes 26 mm three-layer cross-laminated timber (CLT) according to [DIN \(2022\) EN 13353](#) (Figure [2.1](#)).

The assembly integrates three core components. Vertical U-shaped carriers with semi-circular cutouts form a form-fit cradle for the drum, ensuring uniform weight distribution and preventing vessel rotation under high torque. A longitudinal support and side wall tie these carriers together, providing the necessary rigidity against longitudinal tipping while serving as a mounting interface for the water distribution system. Finally, a thick base plate expands the system footprint and lowers the center of gravity.

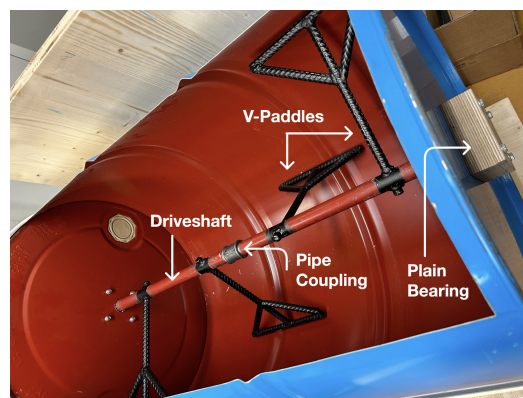
A stability test measuring tilting moments along the radial and longitudinal axes validated the structural integrity. The design is technically sound because the stability factor (S)—defined as the ratio of the restoring moment to the tilting moment—exceeded 1.0 ([Bender & Göhlich, 2020a](#), p. 241). This empirical verification confirms that the panel-based geometry withstands the mechanical stresses of manual substrate processing.

2.2.3 Mixing: Agitator Assembly

The agitator is designed to achieve a uniform mixture of heterogeneous agricultural residues, specifically cotton husks (1 to 2 cm) and fine sawdust, with a target moisture content of 60 to 70 wt % ([Kisseka, 2026](#)) as required in the research questions 1 and 2 ([1.4](#)).



(a) Model agitator (scale 1:5)



(b) Final full-scale agitator

Figure 2.2. Model and final agitator configuration.

1:5 scale agitator model equipped with a T-paddle configuration (2.2a, left). Final full-scale agitator (2.2b, right)

Driveshaft and Hand crank

The central drive component transitioned from an initial monolithic concept to a two-piece DIN (2007) EN 10255 3/4" steel pipe configuration. This modularity allows installation into the rigid mixing vessel. Although sections are joined through standard threaded pipe couplers DIN (2000) EN 10241, initial testing revealed that the couplings alone were insufficient to withstand the high mechanical torques encountered during substrate agitation. To ensure a rigid and durable connection, the interface was reinforced by installing bolts (DIN (2004) EN ISO 4762) that were threaded through holes in both the couplings and the pipes.

Manual power is delivered via a hand crank attached to the driveshaft using threaded pipe couplers (DIN (2000) EN 10241). The lever arm length was maximized to reduce the required input force through mechanical advantage, while remaining within the ergonomic constraints necessary for a standing operator to maintain a consistent rotational frequency.

Agitator Geometry

The selection of the internal agitator geometry was based on a comparative analysis between ribbon blenders and paddle mixers, which according to Oei (2003, pp. 22-23) are both used for the preparation of substrate. While ribbon blenders are effective for blending dry powders through high-shear forces (JM, 2026), a paddle mixer configuration was identified as technically superior for this application.

Paddle-style agitators are specially designed to scoop, lift and tumble in a gentle but thorough mixing action, and are ideal for mixing solids or liquids of various particle size, density, and viscosity (Marion, 2024). Moreover, JM (2026) states that unlike high-shear ribbon blenders, paddle mixers require lower input torque, which is critical for a hand-driven device to mini-

mize operator fatigue. Furthermore, the geometric simplicity of the paddles facilitates rapid cleaning (JM, 2026) and straightforward fabrication using standard welding techniques.

The geometry was refined through 1:5 scale prototyping as shown in figure 2.2a, comparing T-paddle and V-paddle geometries. Although both modeled configurations achieved spatial homogeneity within a few crankshaft revolutions, the V-paddle was selected for its slightly superior longitudinal material transport and enhanced structural stability provided by the V-shaped bracing. In the final implementation, the "V-wings", constructed from DIN (2024) 488-6 Ø 15 mm reinforcing bars, were welded to the driveshaft as shown in figure 2.2b.

Plain Bearings

The driveshaft is supported by wooden radial plain bearings. Although standardized industrial ball bearings were initially considered, they were deemed unsuitable due to their high costs and the difficulty of acquisition in resource-constrained environments. In contrast, wooden plain bearings offer a cost-effective, easy-to-manufacture, and easily replaceable alternative. Inspired by the long-term reliability of traditional spinning wheels, this steel-on-wood interface is highly effective for the low-frequency agitation of around 15 rpm required by the process. The bores in the steel drum are intentionally over-sized to ensure that the shaft is supported exclusively by the wooden bearings, thus preventing inefficient metal-on-metal contact between the pipe and the barrel wall. In addition, the use of grease is recommended to optimize tribological properties and minimize frictional losses and noise. One plain bearing is visible in figure 2.2b.

2.2.4 Watering: Manifold System

Hydrostatic Reservoir

The irrigation system is designed to achieve a target substrate water content of 60–70 wt % as seen in research question 2 on chapter 1.4. To avoid the complexity of mechanical pumping, the design utilizes the hydrostatic pressure generated by a high-elevation reservoir connected to a distribution manifold through a ISO (2022) 8330 PVC spiral suction hose. The reservoir consists of a 20-liter standardized industrial plastic canister (WERIT, 2026) equipped with a discharge tap. This canister is positioned on a wooden stand integrated into the supporting structure at a height of 170 cm. This elevation maximizes the hydrostatic head available while remaining within the ergonomic limits for manual handling. To ensure structural integrity, the stand is reinforced with cross-bracing to counteract static tipping moment and transient torques resulting from manual canister placement or water sloshing. The assembled water distribution system is visible in figure 2.1.

Manifold

The distribution manifold comprises a 3/4" PE pipe complying with [DIN \(2025\)](#) EN 12201, featuring 11 discrete discharge orifices. Water enters the left terminal at a volumetric flow rate of $Q_{\text{in}} = v_{\text{in}} \cdot A_{\text{pipe}}$, while the right end is sealed with a threaded cap ([DIN \(2000\)](#) EN 10241) to maintain internal pressure. This configuration drives the outflow through the individual nozzles ($Q_i = v_i \cdot A_{\text{nozzle}}$, for $i \in [1; 11]$) across the substrate vessel. The hydraulic design targets a flow velocity variance between the first and eleventh nozzle of less than 10% ($v_{11} \geq 0.9v_1$). Assuming steady, incompressible flow, the hydraulic behavior along the pipe is governed by the extended Bernoulli equation ([Kundu et al., 2016](#), p. 299):

$$\frac{1}{2}\rho v_{\text{in}}^2 + \rho g \Delta h = p_i + \frac{1}{2}\rho v_i^2 + \Delta p_{l,i} = \text{constant} \quad (2.1)$$

where ρ is the fluid density, g is the gravitational acceleration, p_i and v_i are the static pressure and velocity at orifice i , and $\Delta p_{l,i}$ represents the cumulative localized pressure loss induced by continuous mass extraction ([Jenni et al., 2025](#), p. 53). Evaluating this analytical model yields a theoretical ideal orifice diameter of 1.70 mm.

However, empirical testing revealed significant deviations from this idealized model. Orifices with diameters of 2.0 mm or smaller exhibited extreme susceptibility to manufacturing defects and clogging. Drilling burrs and fine particulate matter frequently blocked the openings, causing stochastic flow fluctuations rather than a predictable pressure gradient. Qualitative assessments confirmed that for diameters below 2.0 mm, these structural and environmental blockages completely outweighed the theoretical effects of hydraulic head, nozzle count, and calculated pressure drop.

To overcome these manufacturing limitations and ensure operational reliability, the orifices were systematically enlarged to 3.0 mm. This diameter optimization significantly increases volumetric throughput. The resulting reduction in fluid delivery time shortens the total mixing cycle, which directly minimizes operator labor and enhances the economic viability of the substrate preparation process.

2.2.5 Portioning: Filling Line

The portioning unit is integrated beneath the mixing drum to facilitate the volumetric discharge of substrate into cultivation bags by using the geometric constraints of the frame as integrated mechanical stops and mounting points. The system is designed to accommodate standard plastic bags (7–8 inch diameter) common in the Ugandan market ([Kisseka, 2026](#)). The development of the portioning unit followed an agile methodology, aiming to achieve a volumetric discharge tolerance of $\pm 10\%$, as required in Research Question 3 stated in [1.4](#). Fur-

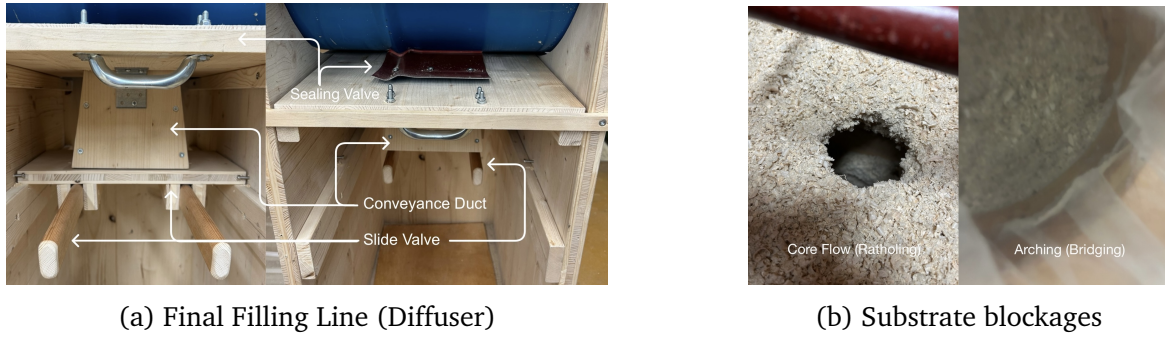


Figure 2.3. Final version of filling line and visualized substrate blockages.

(2.3a, left) Final application of the portioning unit. Sealing valve in open, slide valve in closed (left) and open (right) position. (2.3b, right) Substrate blockages. Core flow (ratholing) inside the mixing vessel (left), bridging (arching) within the conveyance duct (right).

thermore, material loss should be kept to a minimum when transferring the substrate from the mixing vessel to the bags. As described in the following, the filling line comprises three functional stages, an upper sealing valve for process containment, a conveyance duct serving as a volumetric measuring chamber, and a lower manual slide valve for discharge control.

Sealing Valve

Positioned at the upper interface between the mixing vessel and the portioning unit, this component ensures a form-fit seal during the mixing and moistening phases to prevent premature discharge. Once the mixing process is complete, the valve is manually retracted to allow the substrate to descend into the conveyance duct via gravity.

Conveyance Duct: overcoming substrate blockages

This central component serves as the volumetric measuring chamber, as its internal geometry defines the target portion size. In addition, it acts as the structural conduit between the upper sealing mechanism and the lower discharge interface (slide valve).

The development of the conveyance duct followed an iterative process to overcome significant flow obstructions inherent to the mechanics of bulk solids. The initial prototype, featuring a quadratic funnel geometry constructed from wood with a large intake at the top narrowing towards the outlet ($A_{\text{top}} > A_{\text{bottom}}$), resulted in a failure as the majority of the substrate remained lodged within the funnel as imaged in figure 2.3b. According to Stuess (2009, pp. 462-464), flow is inhibited when the shear strength of the bulk solid exceeds the total weight of the substrate above the discharge opening. The angle of the tapered walls of the funnel geometry, as well as the roughness of the wood surface, were determined to be the primary causes of substrate entrapment (Stuess, 2009, p. 470). Moreover, the quadratic cross-section

promoted stagnant zones, leading additionally to the formation of stable arches within the substrate (Stiess, 2009, p. 469).

To mitigate friction and decrease the funnel's cone angle, a second iteration utilized a cylindrical PE-pipe with a smooth surface. Although effective for dry material, the moist substrate exhibited significant bridging after flow interruption through the slide valve in its closed position. This phenomenon is categorized as time consolidation, where the adhesive bonds between particles strengthen during the static phase at the slide valve interface (Stiess, 2009, pp.464). According to Bender and Göhlich (2020b, pp. 528), below the minimal critical cross-section, discharge aids are mandatory because gravity flow cannot be assumed. Therefore, a subsequent attempt to use a flexible ventilation hose as a manual discharge aid was validated, but proved to be ineffective ergonomically and was also discarded due to limited local material availability (Kisseka, 2026).

To definitively address these mechanical bottlenecks, the final design implements a divergent quadratic diffuser geometry ($A_{\text{top}} < A_{\text{bottom}}$) shown in figure 2.3a. The divergent geometry ensures that the substrate expands as it descends, successfully preventing blockages even after the consolidation phase at the valve interface.

Slide Valve: Bagging Interface and Attachment Optimization

The commercial viability of the portioning system is primarily dictated by its operational throughput. If the implemented bagging process exceeds the duration of portioning purely by hand, the mechanical system loses its technical justification. Consequently, the design of the bag attachment interface underwent an iterative optimization process focused on reducing cycle times and ensuring material integrity.

The initial design utilized a circular aperture in which plastic bags were inserted and secured between two clamping plates. To prevent slippage under the weight of the substrate, sharp fasteners were arranged circumferential to perforate and anchor the bag. Although functionally reliable in terms of load carrying capacity, clamping systems were discarded due to their low ergonomic efficiency and structural damage to the bags. The final bagging interface as seen in figure 2.3a utilizes a parallel rod suspension system. This easily accessible configuration inspired by a clothes drying rack allows the operator to mount and fill the bags in a single fluid motion and significantly reduces the overall cycle time compared to the initial circular clamping mechanisms.

3 RESULTS AND DISCUSSION

3.1 Deliverables and Operational Throughput

The primary physical deliverable of this engineering project is a fully functional, full-scale prototype optimized for automated substrate hydration, mechanical mixing, and precise portioning. To ensure reproducibility and facilitate local manufacturing, a complete set of open-source engineering design drawings, manufacturing schematics, and user manuals has been uploaded to a [GitHub repository](#).

Although scientific evaluation focuses on the quality of the substrate preparation process, the primary performance indicators for target end-users like *Hello Mushroom U Ltd* are maximum batch volume capacity and minimal total processing cycle time ([Kisseka, 2026](#)). The finalized prototype demonstrates that the mechanical mixing phase is completed within 5 minutes. The subsequent portioning and bagging sequence takes around 10 minutes. This represents a substantial optimization compared to the current manual baselines. According to an initial estimate by [Kisseka \(2026\)](#), the new apparatus could reduce the workload by more than half. The batch size remains limited by the standardized oil drums used as mixing vessels, but the rapid cycle time effectively compensates for this volume constraint.

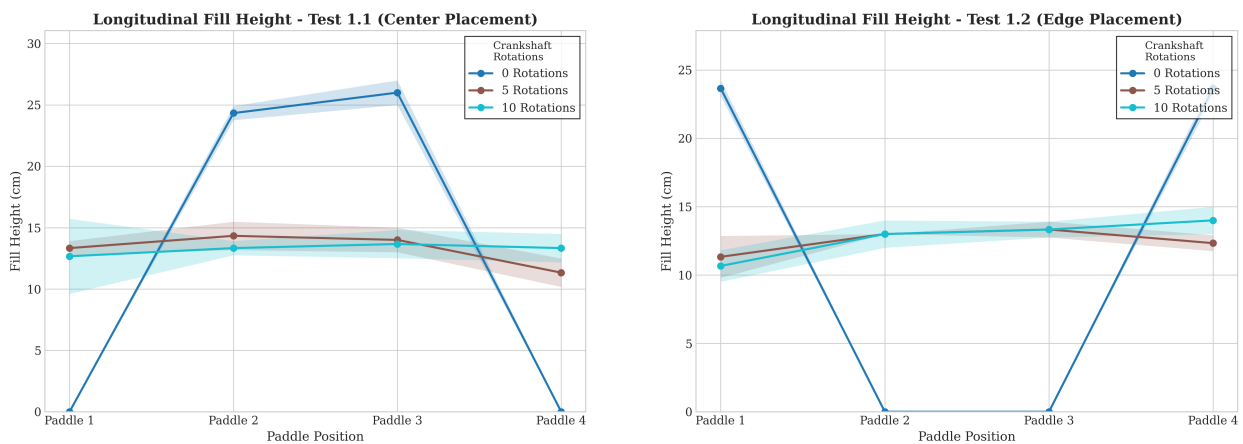
3.2 Evaluation of Substrate Homogeneity (RQ1)

3.2.1 Water Content in Bags

To verify the homogeneity of the processed substrate required in research question 1 (1.4), the final water content within the filled and portioned bags was quantified. The spatial moisture distribution among the output portions remained highly uniform throughout the evaluated experiments. A first trial with $n = 5$ probes yielded a standard deviation of $\sigma_{w,bags,2} \pm SEM_2 = (0.93 \pm 0.42)\%$ within the water content inside the bags w_{bags} , using the standard error of the mean ($SEM = \sigma_w / \sqrt{n}$). Similarly, the subsequent run ($n = 5$) confirmed this high reproducibility, demonstrating a standard deviation of $\sigma_{w,bags,3} \pm SEM_3 = (0.62 \pm 0.28)\%$. This

narrow statistical variance with standard deviations below 1% demonstrates that the internal water distribution becomes very homogeneous during the process and therefore satisfies research question 1 of 1.4.

3.2.2 Longitudinal Transport Efficiency of Agitator



(a) Fill Level Adjustment (Center Placement)

(b) Fill Level Adjustment (Edge Placement)

Figure 3.1. Longitudinal transport efficiency of the agitator under non-uniform initial loading profiles.

Data lines represent the mean values of three independent trials recorded after different agitator rotations. Shaded regions denote error bands calculated from the standard deviation.

To evaluate the longitudinal transport efficiency of the agitator, experiments were conducted using non-uniform initial material distributions. The substrate was loaded either as a centralized pile with empty margins (Figure 3.1a) or concentrated exclusively at the outer boundaries with an empty center (Figure 3.1b). The geometric equalization of the material profile was subsequently monitored as a function of successive agitator rotations. The empirical results demonstrate rapid, bidirectional longitudinal transport both outward from the center and inward from the boundaries. The spatial fill levels are equalized efficiently within the first few cycles; after five rotations, variations in substrate height decrease toward a neglectable threshold. The narrow distribution of the error bands across the three independent trials underscores the high reproducibility of this material transport behavior. Depending on the angular position of the crankshaft, individual paddles engage with the substrate at varying depths. This localized displacement creates a transient shoveling effect that induces minor fluctuations in the local fill height, complicating precise instantaneous quantification and introducing a measurement uncertainty. Nonetheless, the macroscopic trend toward rapid, uniform material distribution remains robust and clearly visible.

3.3 Evaluation of Uniform Hydration (RQ2)

3.3.1 Manifold Flow Distribution

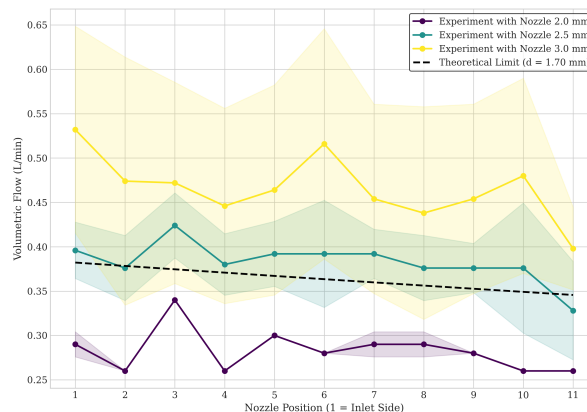


Figure 3.2. Manifold flow distribution.

Comparison of different nozzle diameters against the theoretical hydraulic limit. Solid lines represent the empirical mean values across all trials, while the shaded regions denote error bands.

As observed during the technical design phase (Section 2.2.4), manufacturing inaccuracies dominate the fluid behavior at or below nozzle diameters of 2.0 mm. This phenomenon is clearly reflected in the empirical profiles shown in Figure 3.2. Furthermore, the dataset for the 2.0 mm configuration carries lower statistical significance because only two experimental trials were performed due to the aforementioned fabrication challenges. In contrast, the discharge profiles for the 2.5 mm nozzle configuration align closely with the theoretical hydraulic limit calculated using the Bernoulli equation (2.1). The relative standard deviation for this 2.5 mm configuration is $RSD_{2.5} = (10.87 \pm 2.46)\%$, which tracks only marginally above the target uniformity threshold of $\pm 10\%$ required in research question 2 (1.4). The relative standard deviation (RSD) and the corresponding error $\Delta(RSD)$ are calculated using Gaussian error propagation $RSD \pm \Delta(RSD) = \frac{\sigma}{\mu} \pm \Delta(\frac{\sigma}{\mu})$, with standard deviation σ and mean μ .

The subsequent enlargement of the nozzle bore to 3.0 mm aimed to optimize processing speed, as discussed in 2.2.4. This adaptation is justified by a distinct increase in the total cumulative discharge rate from 4.21 L/min in the 2.5 mm bore configuration to 5.13 L/min through the 3.0 mm nozzles. However, the 3.0 mm bores did not meet the specified uniformity criterion, resulting in a high relative deviation of $RSD_{3.0} = (23.75 \pm 5.58)\%$ between the individual nozzles. The expanded error bands for the 3.0 mm configuration in Figure 3.2 underscore a reduction in flow predictability under higher volumetric flow regimes.

Nevertheless, the non-uniform flow inherent in the finalized 3.0 mm bores represents an acceptable engineering trade-off between dosing precision and operational velocity. Crucially, the

distribution discrepancies introduced by the manifold system (3.3.1) are dynamically mitigated and fully compensated for by the highly efficient longitudinal material transport performed by the agitator (3.2.2) as will be seen in the next section 3.3.2.

3.3.2 Water content while Mixing

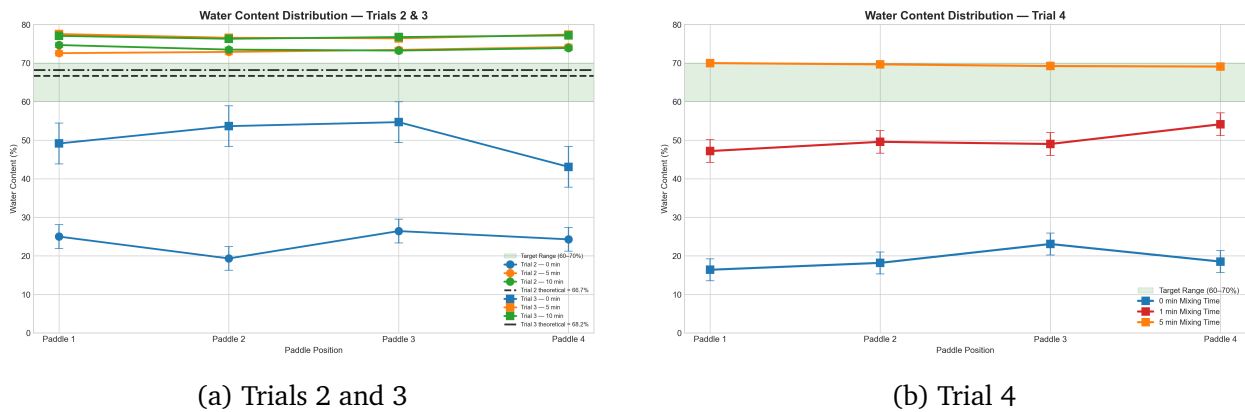


Figure 3.3. Substrate moisture content profiles recorded across successive mixing intervals. Data for substrate samples which were extracted after different time periods of mixing.

To address research question 2 of Subchapter 1.4, the hydration and homogenization performance of the device was verified by extracting substrate core samples at each paddle position after specified mixing durations to analyze differences in moisture content. The resulting spatial moisture distributions are presented in Figure 3.3. The empirical results indicate that the final substrate moisture content is heavily influenced by the baseline moisture level of the initial substrate ingredients. The evident overshoot in final water content above the anticipated target range in Trials 2 and 3 as visible in Figure 3.3a is directly attributable to an underestimation of this baseline parameter. Specifically, the raw material used in Trial 2 possessed an initial mean water content of $\overline{w}_0 \pm \sigma_{w_0} = (23.8 \pm 3.08)\%$, while the substrate in Trial 3 exhibited an initial value of $\overline{w}_0 \pm \sigma_{w_0} = (50.1 \pm 5.28)\%$. Because these high initial moisture levels were undocumented prior to the experiments, the pre-calculated theoretical targets visible as dashed lines in Figure 3.3a settled below the actual empirical outcomes.

Crucially, this over-hydration stems from an initial parameter configuration error rather than a mechanical limitation of the apparatus. The mixing system demonstrates excellent moisture distribution efficiency. Aggregating the metrics after 5 minutes of mixing yields an average absolute deviation in water content of $\sigma_w \pm \text{SEM} = (0.45 \pm 0.06)\%$ ($n = 12$). Maintaining a deviation below 1.0% signifies exceptionally high blending precision and confirms uniform moisture distribution throughout the vessel.

As illustrated by the virtually overlapped moisture profiles at the 5- and 10-minute marks in

Figure 3.3a, the spatial deviations at the paddle positions are already negligible immediately after irrigation completion. The homogenization capability is further substantiated by the finalized Trial 4 run (Figure 3.3b), which achieved a highly stable mean water content ($\bar{w}$) of 69.5% with a minor standard deviation (σ_w) of $\pm 0.41\%$. This output falls within the optimal 60–70% operating threshold required to satisfy Research Question 2 (1.4) and is validated in the average water content in the finally portioned bags of $\bar{w}_{\text{bags}} \pm \sigma_{w,\text{bags}} = (69.55 \pm 0.62)\%$.

3.4 Evaluation of Portioning Accuracy (RQ3)

Table 3.1. Portioning performance summary

Portioning performance mean portion mass, standard deviation to the mean (SD), relative standard deviation (RSD), and material loss per fully portioned batch.

Trial	Prototype	Mean (kg)	SD (kg)	RSD (%)	Loss (wt%)	Note
1	PE-Pipe	0.414	0.153	36.89	45.42	all bags
2	Diffuser	1.367	0.105	7.69	8.74	all bags
3	Diffuser	1.146	0.186	16.25	3.24	all bags
3	Diffuser	1.235	0.105	8.52	—	bags 6–18

The functional efficiency of the portioning mechanism is heavily dictated by the geometric configuration and material properties of the conveyance duct as already observed during the product development phase discussed in 2.2.5. This dependency is demonstrated by the performance collapse of the PE-pipe configuration visible in Table 3.1. This setup induced severe flow obstructions, resulting in an unacceptably high relative standard deviation of $\text{RSD}_1 = (36.89 \pm 4.70)\%$ calculated using Gauss error propagation introduced in 3.3.1, and a material loss in the whole batch of around 45%. These quantitative deficiencies directly substantiate the structural arching and bridging dynamics analyzed in Section 2.2.5.

Despite geometric improvements in the subsequent iteration, the portioning system remains highly sensitive to operator consistency. However, when the filling procedure is executed in accordance with operational specifications, the system delivers the precision required to satisfy Research Question 3 defined in chapter 1.4. As documented in Table 3.1, the refined prototype utilizing a diffuser achieved a relative standard deviation of $\text{RSD}_2 = (7.69 \pm 1.25)\%$ in Trial 2 computed accordingly to RSD_1 described above. Furthermore, when adjusting for initial user errors by isolating bags 6–18 in Trial 3, the mass variation was restricted to $(8.52 \pm 1.75)\%$ in terms of $\text{RSD}_{3,\text{annotated}}$. Both configurations successfully fulfill the $\pm 10\%$ tolerance threshold mandated by Research Question 3. Additionally, disciplined execution minimizes material waste, reducing total batch-mass losses to a marginal level of around 3% as demonstrated in Trial 3 shown in Table 3.1.

4 CONCLUSIONS

4.1 Prototype Performance and Research Questions

This thesis developed and evaluated a low-cost, hand-driven substrate processing device. The system reduces manual labor in smallholder mushroom cultivation in Kampala, Uganda. It combines mechanical homogenization, hydrostatic hydration, and volumetric dosing into a single mechanism. Experimental validation confirms that this low-tech design, built from universally available materials, offers a reliable alternative to capital-intensive machinery. The design successfully addresses the overarching research objectives.

Prototype testing provides direct answers to the research questions in 1.4. Regarding substrate homogeneity (**RQ1**) and uniform hydration (**RQ2**), the modular V-paddle agitator and hydrostatic manifold performed exceptionally. 5-minute mixing phases yield an average absolute moisture deviation of $0.45\% \pm 0.06\%$. This uniformity transferred effectively to the cultivation bags. Final bag moisture standard deviations were $\sigma_{w,bags,2} = 0.93\% \pm 0.42\%$ and $\sigma_{w,bags,3} = 0.62\% \pm 0.28\%$. Both values lie within the target range of 60–70 wt%.

The filling line satisfies the requirements for volumetric portioning accuracy (**RQ3**). The conveyance line uses a divergent quadratic diffuser and mechanical slide valves. During the 10-to-15-minute bagging sequence, the system minimized material loss and kept mass deviations well below the $\pm 10\text{wt}\%$ tolerance threshold. Compared to manual processing, the device substantially reduces labor hours.

4.2 Limitations and System Boundaries

Operational boundaries and technical limitations constrain the current prototype. The maximum batch capacity cannot be defined by mass due to highly variable residue bulk densities. Instead, the system is volume-limited. Blending requires that raw materials do not exceed half of the geometric filling height to prevent stagnation zones. For the 210 L drum, this restricts the maximum operational volume to approximately 100 L. Additionally, the direct, non-g geared manual drive requires substantial physical exertion, imposing ergonomic limits on continuous

operation.

Several long-term reliability and contextual aspects remain unresolved. Experimental validation was restricted to immediate post-fabrication performance. Consequently, structural durability under cyclic loading and heavy-duty conditions remains unquantified. Prolonged exposure to high humidity, abrasive residues, and mechanical stress may degrade the lacquered wooden frame and accelerate steel agitator corrosion. Finally, local manufacturing feasibility, component supply chains, and long-term user acceptance in Uganda require further field verification.

4.3 Recommendations for Future Engineering Work

Future research must prioritize extended field validation under real operating conditions in Uganda to address these technical gaps. Long-term durability trials are essential to analyze structural wear profiles and monitor anti-corrosion coating degradation over hundreds of processing cycles. To optimize the physical interface and reduce operator fatigue, the integration of a mechanical gearbox into the drive assembly should be investigated, alongside the potential development of a high-stiffness steel frame to replace the moisture-sensitive wooden structure. Finally, characterizing the mechanical flow behavior of a wider spectrum of residue particle geometries will allow for a precise empirical definition of the operational envelope of the agitator and filling line and their anti-blocking mechanisms.

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APPENDIX A - TEST REPORTS FROM ZERO-SERIES EXPERIMENTS

The documents linked in this appendix contain all test reports from the Zero-Series experiments (Test IDs 00.YY) as described in README.md ([Google Drive](#), [GitHub](#)). These early-stage prototype and component-level tests were conducted during the agile development phase of the substrate mixer and portioning device. Because the data volume of these experiments is low and primarily qualitative in nature, the results are archived as human-readable PDF reports rather than machine-readable datasets.

The Zero-Series experiments include evaluations of structural stability, bearing friction, agitator model performance, and water-distribution orifice behavior. Their purpose was to explore feasibility, identify design weaknesses, and inform the development of the first full-scale prototype. Although these tests are not part of the computational analysis pipeline used for the First-Series experiments (Test IDs 01.YY), they provide essential context for understanding the design decisions and engineering rationale behind the final system.

All reports are stored in the project's [GitHub repository](#) and [Google Drive workspace](#) and are referenced here via persistent hyperlinks to GitHub.

- [00_01_1_1_stability_supportingstructure_testreport.pdf](#)
- [00_02_1_3_stability_friction_plainbearing_testreport.pdf](#)
- [00_03_1_2_agitator_model_testreport.pdf](#)
- [00_04_1_4_orifices_testreport.pdf](#)

APPENDIX B - DATA MANAGEMENT FOR FIRST-SERIES EXPERIMENTS

Introduction

This appendix documents the data-management structure used for all First-Series experiments (Test IDs 01.YY) in [README.md \(GitHub\)](#). These datasets follow a fully reproducible, script-based workflow in accordance with international research-data standards. All raw measurements are stored as machine-readable CSV files, all derived variables are computed programmatically, and all metadata are defined in a central codebook. The complete processing pipeline can be regenerated at any time from the raw data alone.

The following sections provide hyperlinks to the raw and derived datasets, the metadata files, and the Python scripts that implement the computational workflow.

Data Structure and Metadata

- `/data/raw_data/` (All directly measured CSV files. [Google Drive](#), [GitHub](#))
- `/data/derived_data/` (Tidy datasets of processing output. [Google Drive](#), [GitHub](#))
- `/data/metadata/README.md` (Metadata and documentation. [Google Drive](#), [GitHub](#))
- `/analysis/README.md` (Description data processing pipeline. [Google Drive](#), [GitHub](#))
- `/data/metadata/codebook.csv`
(Variable definitions, units, ranges, descriptions. [Google Drive](#), [GitHub](#))
- `/data/metadata/file_list.csv`
(Lists all files of data management. [Google Drive](#), [GitHub](#))

Reproducible Processing Pipeline

The First-Series dataset is processed using two Python scripts. Short excerpts are shown below; full scripts are available via hyperlink.

01_build_derived_data.py

[Download full script, Google Drive](#)

[Download full script, GitHub](#)

```
# 01_build_derived_data.py
# Reads all raw CSV files, reshapes them into long format,
# computes all derived variables, and writes derived_data.csv
# calculates theoretical values
```

01_generate_plots.py

[Download full script, Google Drive](#)

[Download full script, GitHub](#)

```
# 01_generate_plots.py
# Reads derived data csv's and generates all publication-ready figures
# used in the thesis (PNG format, LaTeX-compatible)
```

APPENDIX C - TECHNICAL DRAWINGS, USER MANUAL AND BUILDING INSTRUCTIONS

Introduction

This appendix provides access to all technical drawings, user-manual materials, and building instructions associated with the substrate mixer and portioning device.

The technical drawings define all component geometries used for fabrication, while the user manual and building instructions describe assembly procedures and operational guidelines. A short instructional video is included to support practical implementation in field settings. All files are stored in the project's Google Drive workspace, the GitHub repository or on YouTube and are referenced here via persistent hyperlinks.

Linked Documents

- `/hardware/design/Technical_Drawings_TD`
(Technical drawings of all components, [Google Drive](#), [GitHub](#))
- `/hardware/design/instructions.pdf`
(User manual and building instructions, [Google Drive](#), [GitHub](#))
- `user_manual.mp4`
(Introduction video and user manual. [Google Drive](#), [YouTube](#))

APPENDIX D - CORRESPONDENCES, INDUSTRIAL FIELD VISITS AND EXPERT INTERVIEWS

Introduction

This appendix provides access to all expert interviews, e-mail correspondences, and industrial field-visit reports that were cited in the main thesis and listed in the bibliography. These documents contain qualitative insights that informed the engineering design decisions, contextual understanding of substrate processing practices, and validation of prototype feasibility. All materials are archived in the project's [Google Drive workspace](#) only and are referenced here via hyperlinks. Each entry corresponds directly to a bibliographic citation in the thesis. The Field Visit Reports were taken from Carlos Martin Cazorla's (carlosma@ethz.ch) semester project stored as [2026-msc-sem-proj-carlosma](#) on Google Drive workspace.

Linked Documents

- **Kisseka, S. (2026).** *E-Mail Correspondence with Samson Kisseka from Hello Mushroom U Ltd.*
Company: Hello Mushroom U Ltd., Kampala, Uganda. [Correspondence Transcription](#)
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Bibliography: [Wigger and Fanger, 2026](#)